Suggested problems: 1-49 odd

Problem 1: Describe the curve that is parameterized by

$$x = t \quad y = t^2.$$

Problem 2: Describe the curve that is parameterized by

$$x = -\cos(t) \quad y = \sin(t).$$

Problem 3: Describe the curve that is parameterized by

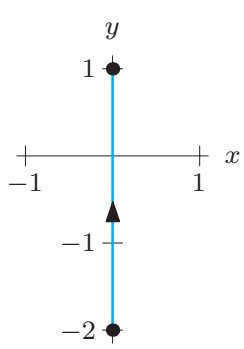
$$x = \cos t \quad y = \sin t \quad z = t.$$

Problem 4: Describe the curve that is parameterized by

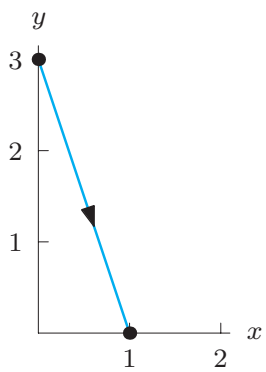
$$x = 2\sin t \quad y = 2\cos t \quad z = -t.$$

Problem 5: Find parameterizations for the following line segments.

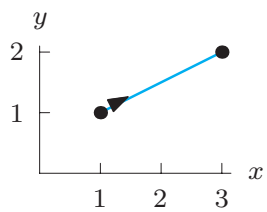
(a)



(b)

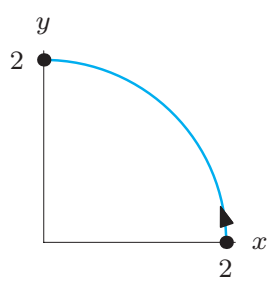


(c)

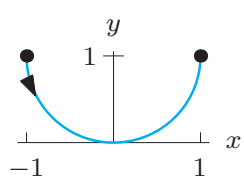


Problem 6: Find parameterizations for the following curves.

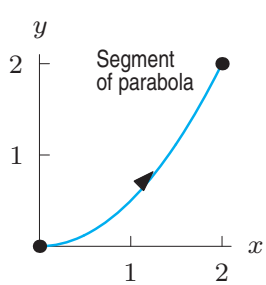
(a)



(b)



(c)



Problem 7: Find a parameterization for the following curves.

(a) A circle of radius 3 centered on the z-axis and lying in the plane $z = 5$.

(b) The circle of radius 2 in the xy-plane centered at the origin, clockwise.

(c) The circle of radius 2 in the xz-plane centered at the origin.

(d) The curve in which the plane $z = 2$ cuts the surface $z = \sqrt{x^2 + y^2}$.

Problem 8: Find a parameterization of the line segment from $P = (2, 5)$ to $Q = (12, 9)$ so that P corresponds to $t = 1$ and Q to $t = 5$.